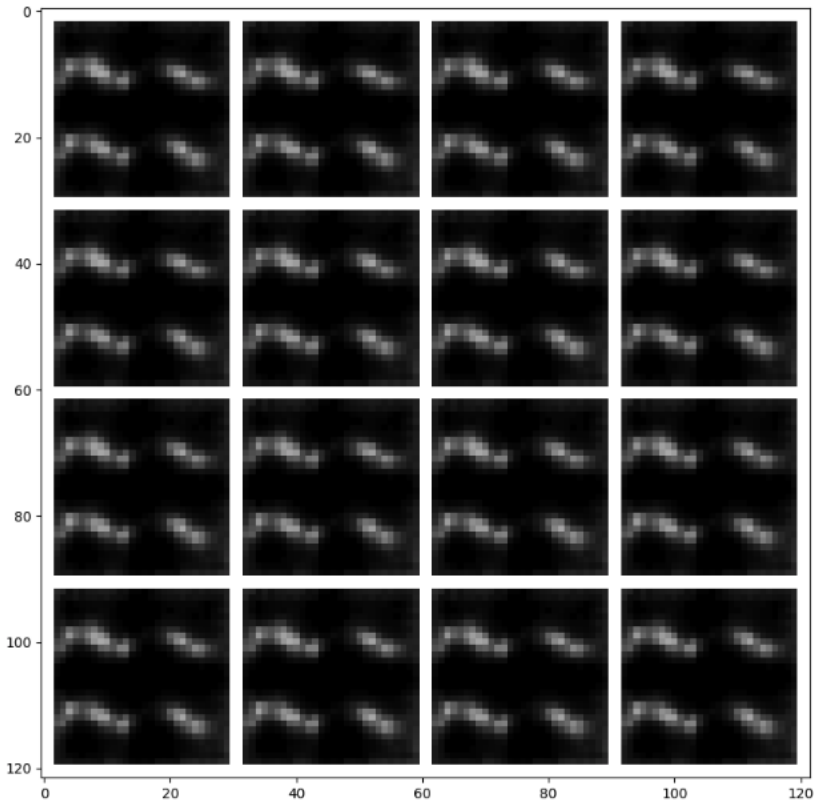
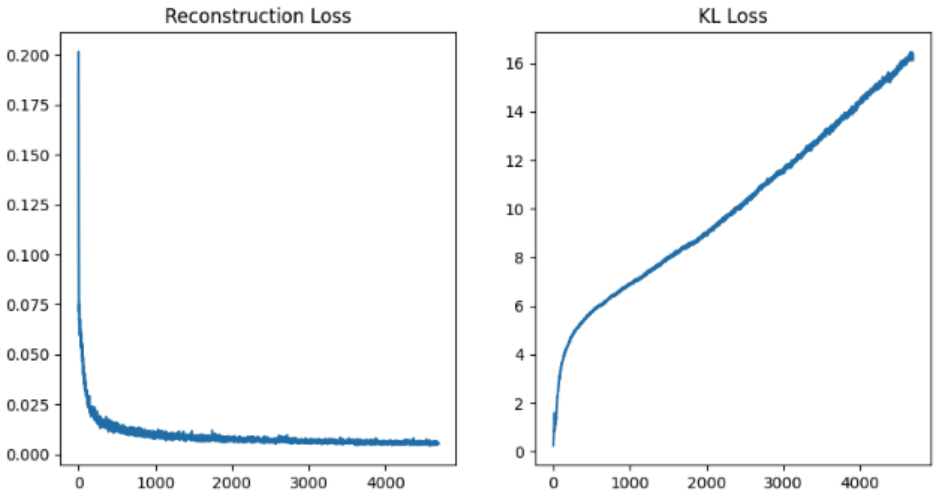
Auto-encoder samples and AE sampling inline question answer.



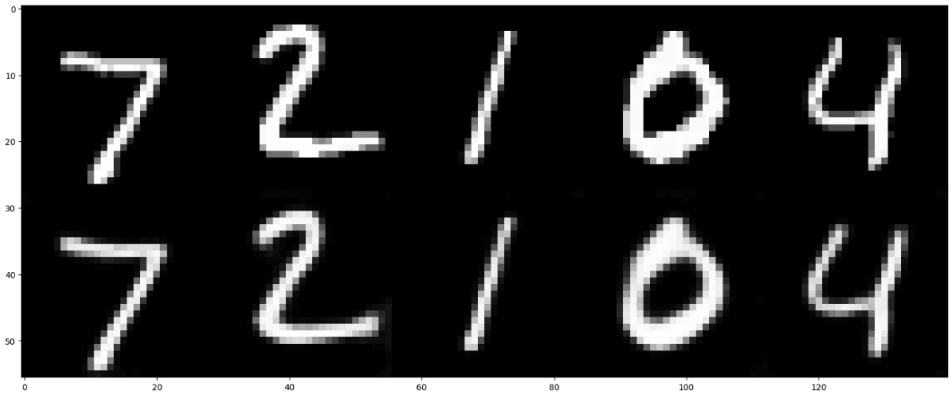
Inline Question: Describe your observations, why do you think they occur? [2pt]
Answer:
The generated samples are very similar to each other and appear blurry, with little diversity. This happens because the auto-encoder does not enforce any structure on the latent space.
During training, the decoder only sees latent codes produced by the encoder. However, at test time, we input latent vectors randomly sampled from a Gaussian distribution. These sampled vectors do not follow the learned distribution.
As a result, many of these Gaussian samples lie outside the regions seen during training, so the decoder cannot map them to meaningful outputs and tends to produce similar blurry images.

VAE training curves, reconstructions and samples

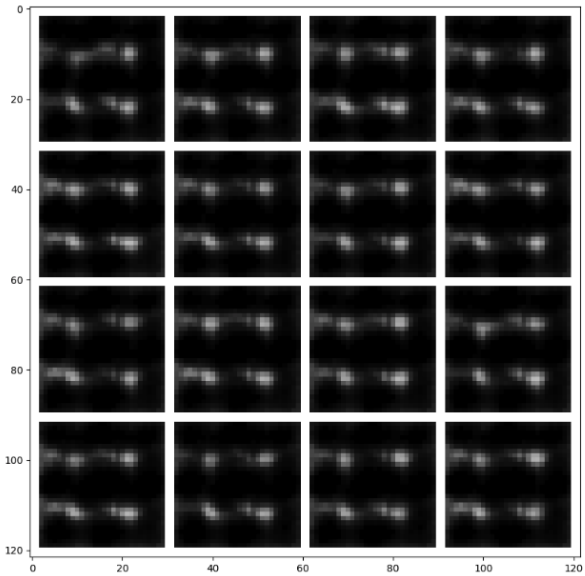
VAE with $\beta = 0$



(a) Training curves (Reconstruction Loss & KL Loss) for $\beta = 0$



(b) Reconstructions for $\beta = 0$ (top: input, bottom: reconstruction)



(c) Generated samples for $\beta = 0$

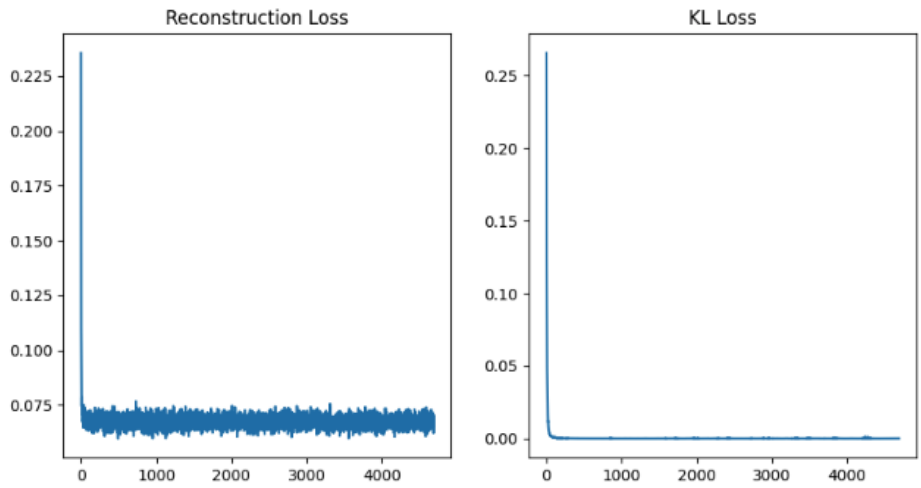
Inline Question: What can you observe when setting $\beta = 0$? Explain your observations! [3pt]

Answer:

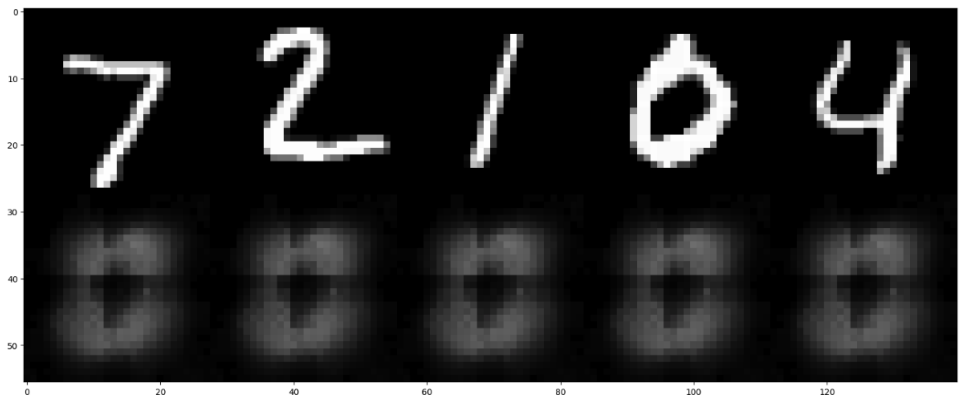
The reconstruction loss decreases rapidly and converges to a very low value, while the KL loss keeps increasing. However, since $\beta = 0$, the KL term does not affect the optimization. The reconstructions are sharp and accurate, and they are almost identical to the input images. This is because $\beta = 0$ removes the KL term, so the model focuses only on reconstruction.

The generated samples are blurry and very similar to each other, with little diversity. Since $\beta = 0$ removes the KL divergence, the latent space is not regularized. As a result, random latent vectors do not correspond to meaningful representations, and the decoder produces similar and blurry outputs.

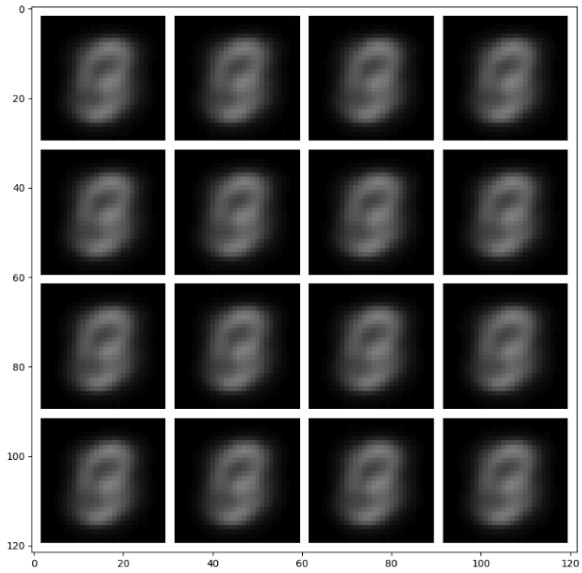
VAE with $\beta = 10$



(a) Training curves for $\beta = 10$



(b) Reconstructions for $\beta = 10$



(c) Generated samples for $\beta = 10$

Inline Question: What can you observe when setting $\beta = 10$? Explain your observations! [3pt]

Answer:

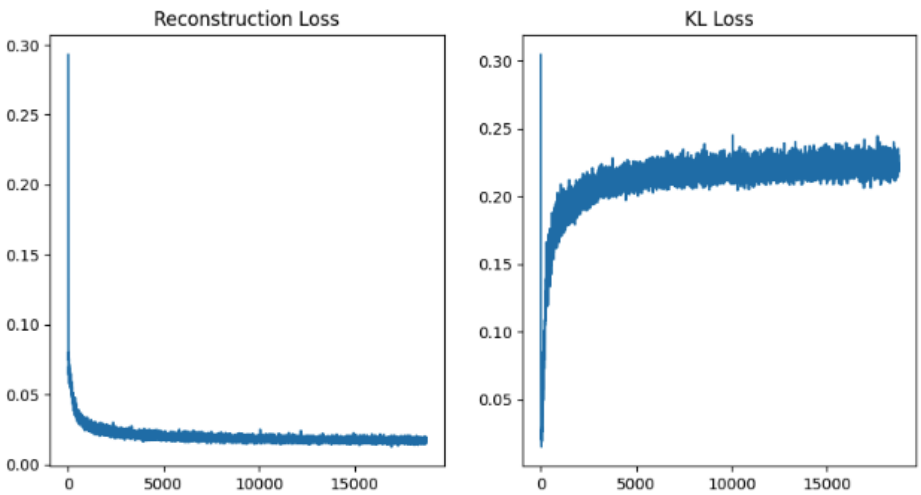
The reconstruction loss is higher than when β is small, while the KL loss quickly becomes close to zero. This is because a large β forces the latent distribution to be close to $\mathcal{N}(0, I)$ for all inputs.

As a result, the latent variables contain very little information about the input, so the reconstructions are blurry and not accurate. The generated samples also look very similar to each other with low diversity, since the latent codes are almost the same regardless of the input.

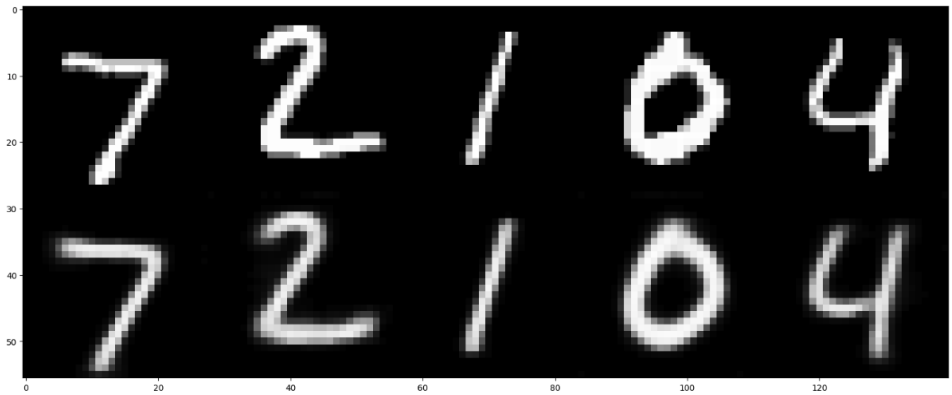
Properties of a Well-Tuned VAE

Inline Question: Characterize what properties you would expect for reconstructions (1pt) and samples (2pt) of a well-tuned VAE! [3pt]
Answer:
A well-tuned VAE should show a good balance between reconstruction and KL regularization.
For reconstructions, the outputs should be close to the input images and preserve the correct digit shapes, although they may appear slightly blurry due to the KL constraint.
For samples, decoding random vectors from $\mathcal{N}(0, I)$ should produce diverse and recognizable digit images. The samples should include different digit classes and show some variation, although they may not be perfectly sharp.
This indicates that the latent space is reasonably well-structured and aligned with the prior, allowing both meaningful reconstructions and diverse samples.

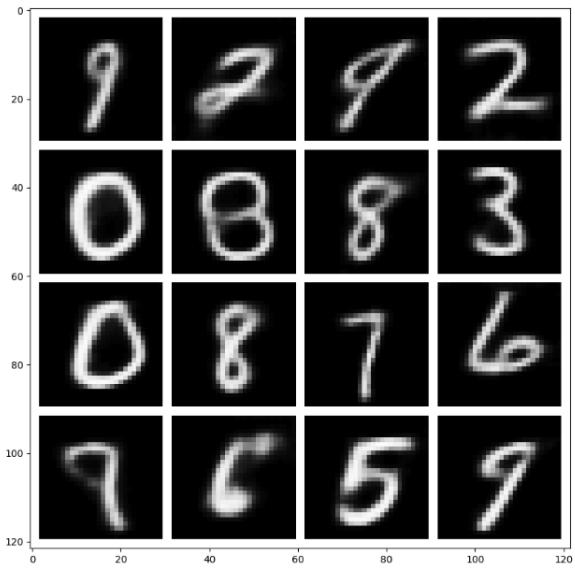
Tuned β Results ($\beta = 0.06$) [5pt]



(a) Training curves



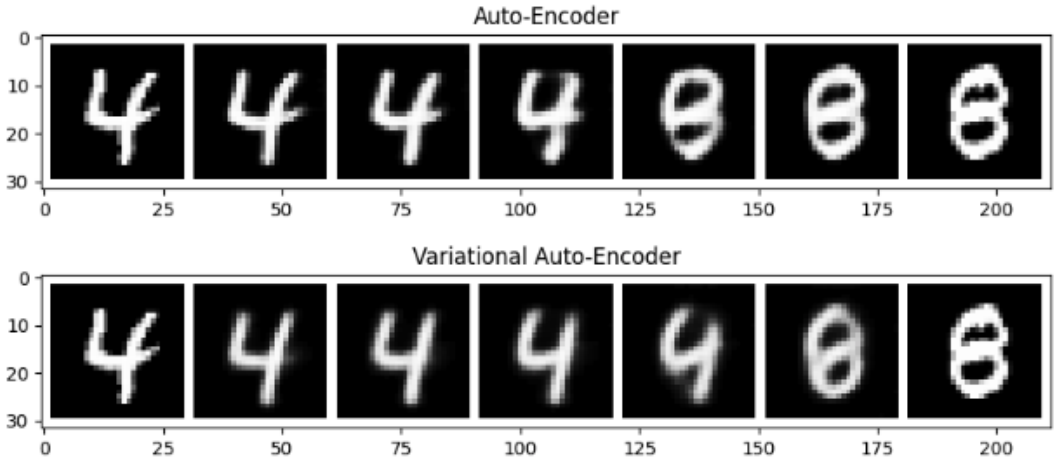
(b) Reconstructions



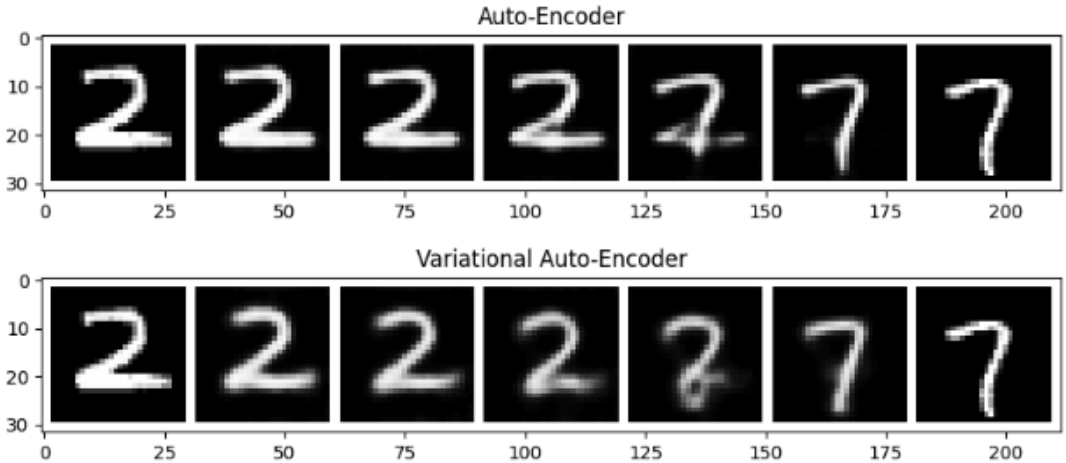
(c) Generated samples

Embedding Space Interpolation [3pt]

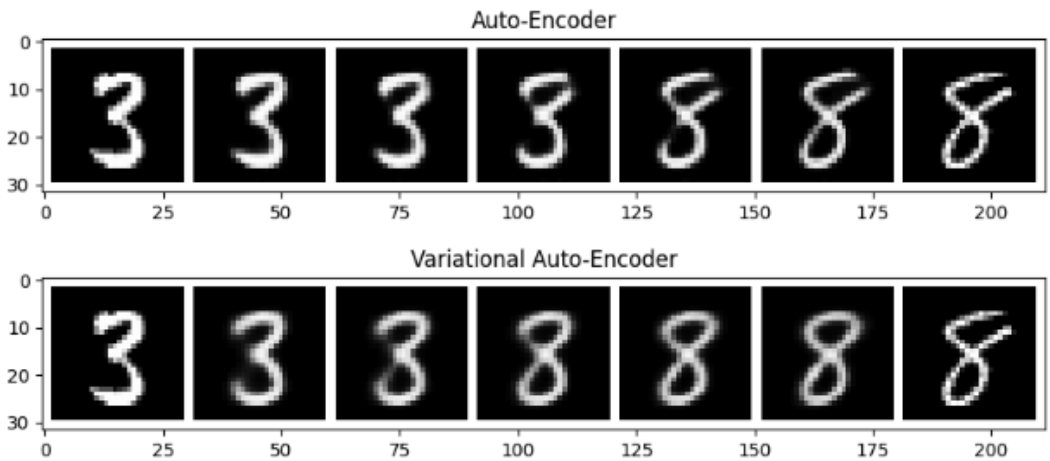
Interpolation: 4 → 8



Interpolation: 2 → 7



Interpolation: 3 → 0



Inline Question: Observations on AE vs VAE Interpolation
Answer:

1. **How do AE and VAE embedding space interpolations differ?**

From the images, the AE shows more abrupt transitions between digits, where intermediate images can look less smooth or slightly inconsistent. This is because the AE does not enforce a structured latent space.
In contrast, the VAE shows more gradual and smooth transitions due to KL regularization, which encourages a well-structured latent space.

2. **How do you expect these differences to affect downstream learning?**

The smooth and structured latent space of the VAE is more useful for downstream tasks. Nearby points in the VAE latent space correspond to similar images, making the representation suitable for tasks such as classification or clustering.
In contrast, the AE's unstructured latent space produces less consistent representations, which are harder to use as meaningful features.